

TED(15)5091

Reg.no.....

REVISION 2015

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FIFTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

FIBRES AND COMPOSITES

(Maximum marks :100)

Time:3hrs.

PART-A

(Maximum marks:10)

I .Answer all questions in one or two sentences .Each question carries 2 marks.

- 1 .Define Sericulture .
2. If 600 g load can break a fibre of denier 20 .Calculate its tenacity.
3. State the raw materials of Kevlar fibre.
4. List two advantages of composites.
5. State the advantages of Hybrid composites. (5*2=10)

PART-B

(Maximum marks :30)

II. Answer any five questions from the following. Each question carries 6 marks.

1. Explain the purification and application of linen fibre.
2. Explain the dry spinning technique of cellulose acetate fibre with a neat diagram.
3. Describe the different types of reinforcement used in composite manufacture.
4. Explain the principle of Centrifugal casting.
5. Explain the processes involved in the manufacture of Wool fibre.
6. Illustrate the technique of Melt spinning process of fibres.
7. Describe the manufacture of PF resin .

(5*6=30)

PART-C

(Maximum marks :60)

Answer one full question from each unit . Each full question carries 15marks.

UNIT-I

- III.a. Explain the manufacture, properties and applications of Coir fibre. 8
- b. List different natural fibers and classify them into vegetable ,animal and mineral fibres. 7

OR

- IV. a Explain the Reeling of Silk fibre with a diagram. 8
- b. Explain staple fibre process for manufacture of Asbestos fibre. 7

UNIT-II

- V.a. Explain the raw materials ,manufacture and properties of Cellulose acetate fibre. 9
- b. List different commercially available forms of Glass fibre and Explain their properties. 6

OR

- VI.a. Mention the spinning technique for Viscose Rayon and explain its manufacture. 8
- b.Explain the manufacture,properties of Polyester Fibre. 7

UNIT-III

- VII.a. Discuss on the Additives used in the curing of Epoxy Resin. 8
- b. Describe the manufacture of UF resin. List its applications. 7

OR

VIII,a. Discuss on curing agents used for epoxy resin curing. 6

b. List the effect of filler on uncured resin and cured resin properties . 9

UNIT-IV

IX.a.Explain the principle and procedure of Vacuum bag moulding . 9

b. List the advantages and disadvantages of Hand layup process . 6

OR

X.a. Explain the manufacture of roofing sheet with emphasis on resin, reinforcement and process selection. 9

b. State the advantages and disadvantages of FRP mould materials-Steel , wood and POP. 6
